

Luke Zhu

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EDUCATION

Harvard University

Cambridge MA, USA

A.B. in Mathematics (Honors) and Physics — Cumulative GPA: 3.99/4.00

Aug 2023 – May 2027

A.M. in Mathematics (Concurrent)

- **Relevant coursework:** MATH55a+b, MATH231a (grad. algebraic topology), MATH232ar (grad. algebraic geometry), MATH136/137 (differential & algebraic geometry) PHYSICS143 (quantum mechanics), CS50 (computer science), STAT110 (probability)

University of Oxford

Oxford, United Kingdom

Visiting Student, Oriel College; Mathematical Institute

Jan 2026 – Jun 2026

- **Relevant coursework:** Part C Master's courses (Computational Algebraic Topology, Networks) alongside additional coursework completed toward Harvard requirements.

PROFESSIONAL EXPERIENCE

Harvard University

Cambridge MA, USA

Course Assistant, MATH101/136

Jan 2025 – present

- Lead weekly office hours and recitation sessions to improve student understanding
- Grade student assignments, helping students improve rigour and exposition

NZPMC Limited

Auckland, New Zealand

Academic Lead

Aug 2022 – present

- Lead academic content design for nationwide mathematics and physics competitions
- Design novel problems requiring physical intuition and creative quantitative reasoning

St Andrew's College

Christchurch, New Zealand

Learning Support & Academic Enrichment Staff, Teacher Aide, Mathematics Tutor

Jan 2023 – Aug 2024

- Designed enrichment materials extending advanced students beyond curriculum
- Mentored students and collaborated with faculty on classroom instruction

HONOURS & AWARDS

John Harvard Scholar: Awarded to students in the top 5% of their Harvard College class (2024, 2025)

Detur Prize: The oldest prize at Harvard College, awarded to sophomores in high academic standing (2025)

Harvard College Research Program: Competitive funding awarded to pursue independent scholarly research (2025)

International Mathematical Modelling Challenge: Won for New Zealand in 2022 (outstanding award).

NZQA New Zealand Outstanding Scholar Award : Recognised among New Zealand's top secondary students for exceptional performance across national scholarship examinations (2023)

PROJECTS AND RESEARCH

Harvard Honors Thesis: Research in quantum stabilizer codes (a mathematical framework for error correction in quantum computers) through algebraic and topological interpretations. Advised by Bowen Yang at Harvard's CMSA.

Affine Group Schemes: Studying affine group schemes and commutative Hopf algebras with a view towards generalising the theory to symmetric monoidal categories. Advised by Professor Kobi Kremnitzer at Oxford.

Machine Learning for Knot Theory: Received Harvard funding to research applications of machine learning in knot theory, with applications to invariant prediction, classification tasks, and biopolymer identification.

HARDMATH 2 (NeurIPS 2025): Developed benchmark tasks for evaluating LLM performance on applied mathematics problems, with emphasis on multi-step modeling, quantitative reasoning, and solution verification.

ETG x NASA VECAR Project: Collaborated with NASA's Volatiles Extraction Cryocapture & Recovery project, designing a lunar-capable device to extract and return soil volatiles to Earth via the Artemis program.

VOLUNTEERING

Harvard Square Homeless Shelter: Volunteer for HSHS on weekends, working breakfast and evening shifts.

Cambridge Math Circle: Teacher for the CMC, a 501(c)3 registered nonprofit that delivers maths enrichment to K-8 students from Cambridge and the Greater Boston area, with a focus on children commonly underrepresented in STEM.